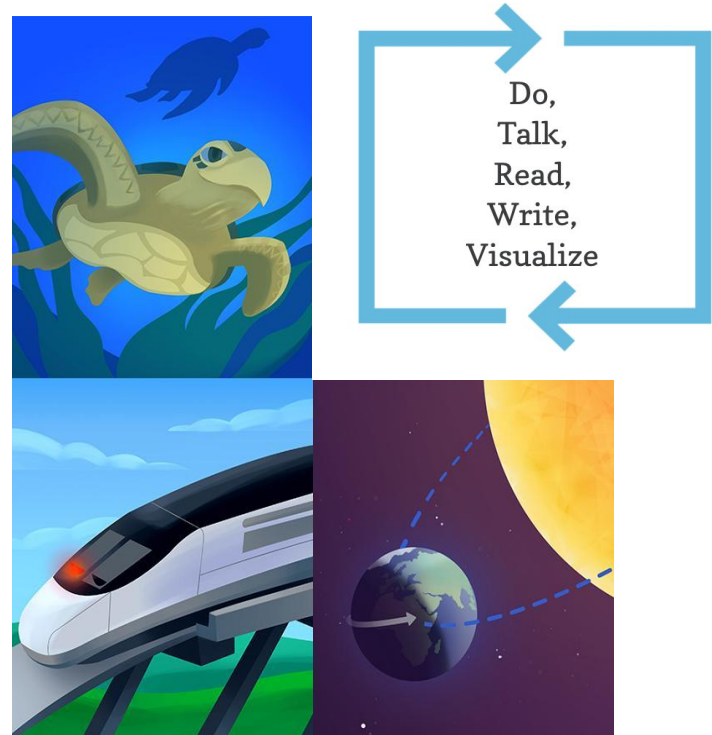
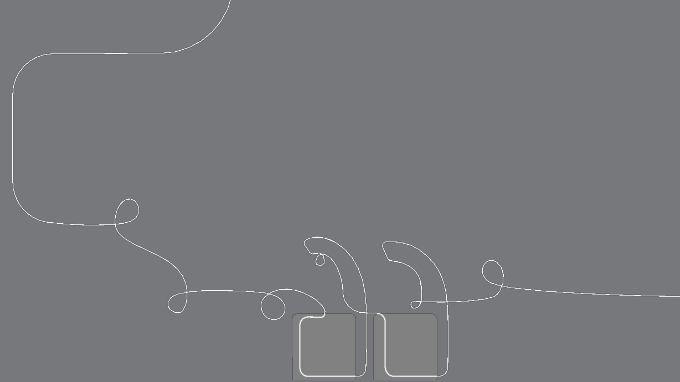


Multimodal, phenomenon-based learning

In each Amplify Science unit, students embody the role of a scientist or engineer to **figure out phenomena**.

They gather evidence from multiple sources, using multiple modalities.

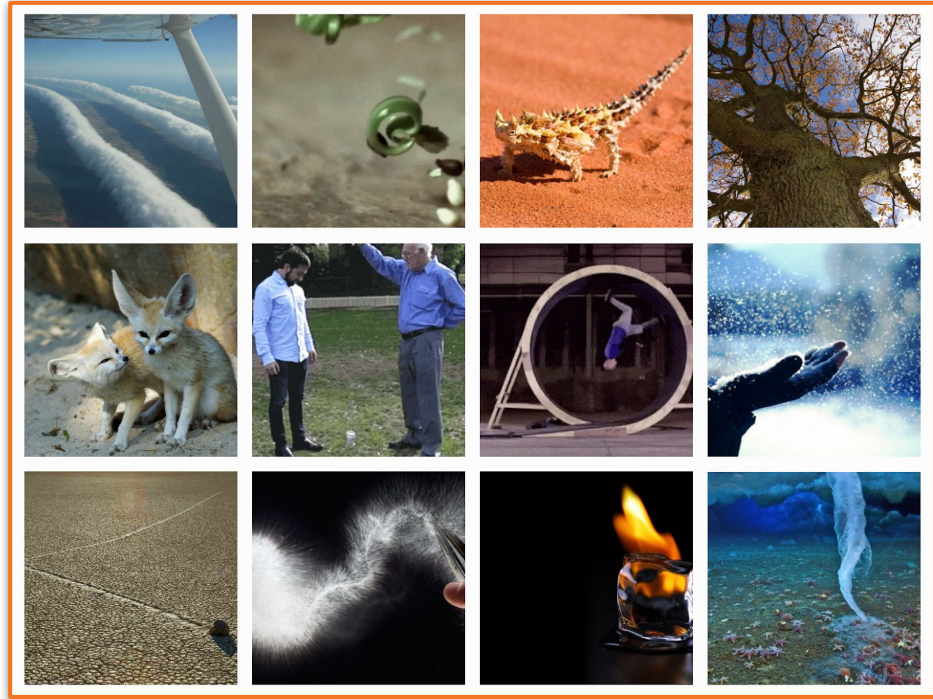




What are PHENOMENA?

Phenomena are observable occurrences.

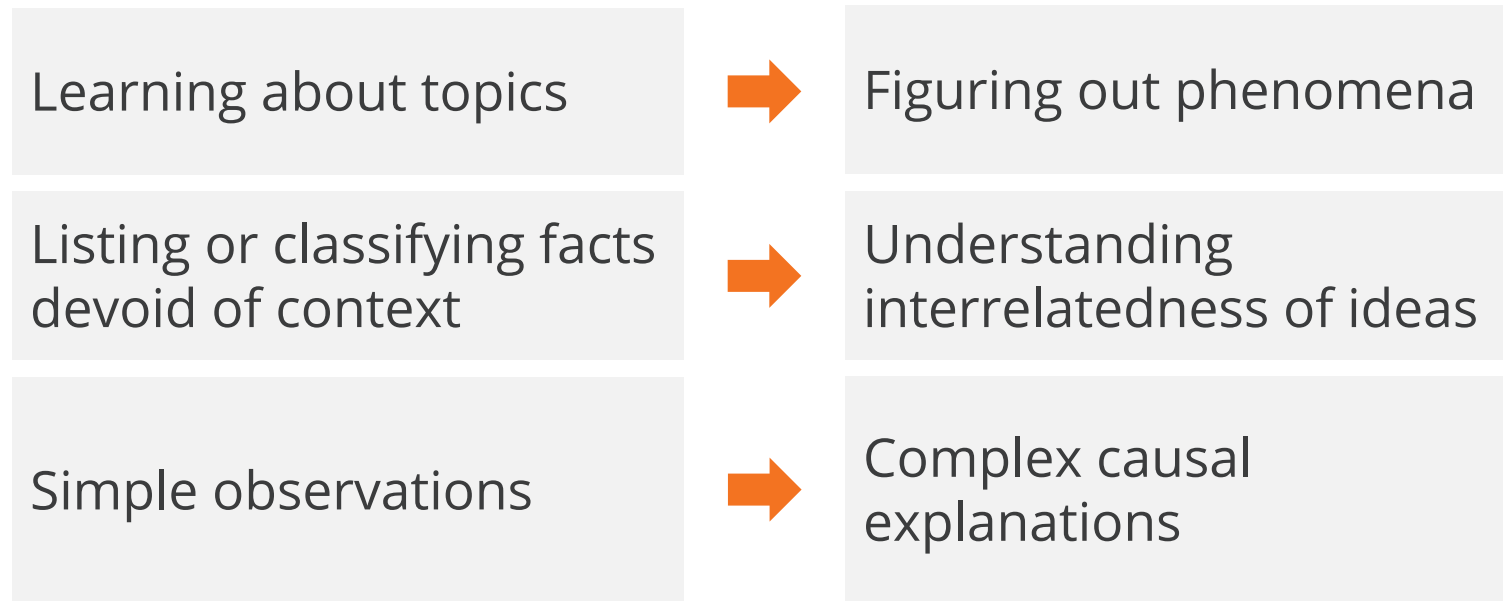
- These occurrences stimulate curiosity or pose problems to for students to solve.
- Students are motivated to ask science questions or design solutions that drive learning.



Amplify Science units focus on phenomena

Topics		Phenomena
all about sea turtles	➡	How do sea turtles defend themselves from sharks?
inheritance and traits	➡	How do organisms get their traits?
ecosystem restoration	➡	How can an ecosystem be restored to its original healthy state?

Shifts called for in the NRC* framework



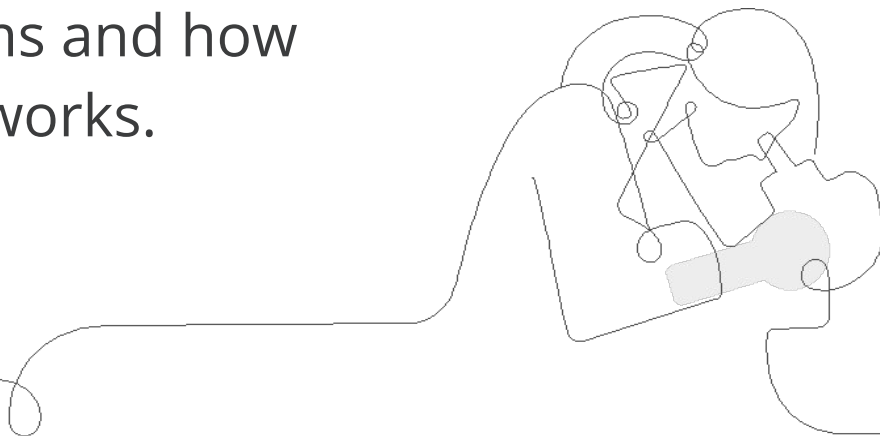
*National Research Council of the National Academy of Sciences, 2011

Figure out, not learn about



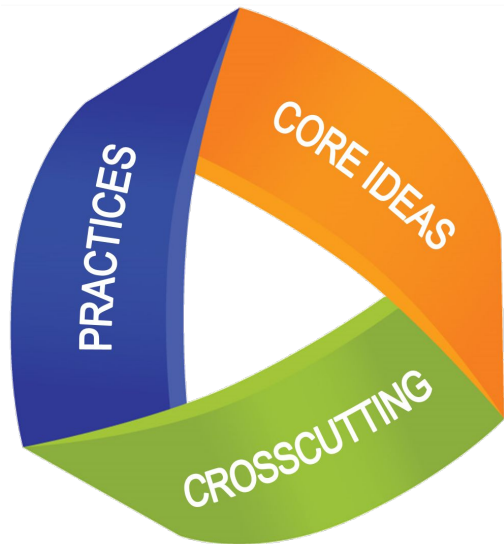
Problem-based deep dives

Students inhabit the roles of scientists and engineers to figure out solutions to real-world problems and how the natural world works.



Three dimensions of NYSSLS

p. 4



Disciplinary Core Ideas

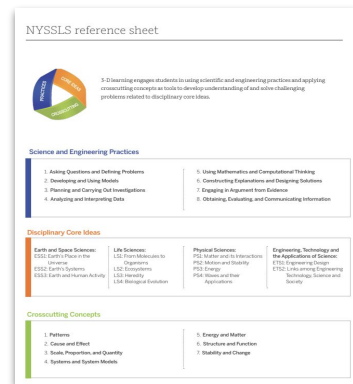
- Describe core ideas in the science discipline (DCI)

Science and Engineering Practices

- Describe behaviors scientists and engineers engage in (SEP)

Crosscutting Concepts

- Describe concepts linking the different domains of science (CCC)



Disciplinary Core Ideas

The foundational concepts lessons are based on

Physical Science

PS1 Matter and Interactions

PS2 Motion and Stability: Forces and Interactions

PS3 Energy

PS4 Wave Properties

Life Science

LS1 From Molecules to Organisms: Structure and Properties

LS2 Ecosystems: Interactions, Energy, and Dynamics

LS3 Heredity: Inheritance and Variation of Traits

LS4 Biological Evolution: Unity and Diversity

Earth & Space Science

ESS1 Earth's Place in the Universe

ESS2 Earth's Systems

ESS3 Earth and Human Activity

Earth & Space Science

ETS1 Earth's Place in the Universe

Science and Engineering Practices (SEP)

How students engage as scientists

1. Asking questions (for science) and defining problems (for engineering)
2. Developing and using models
3. Planning and carrying out investigations
4. Analyzing and interpreting data
5. Using mathematics and computational thinking
6. Constructing explanations (for science) and designing solutions (for engineering)
7. Engaging in argument from evidence
8. Obtaining, evaluating, and communicating information

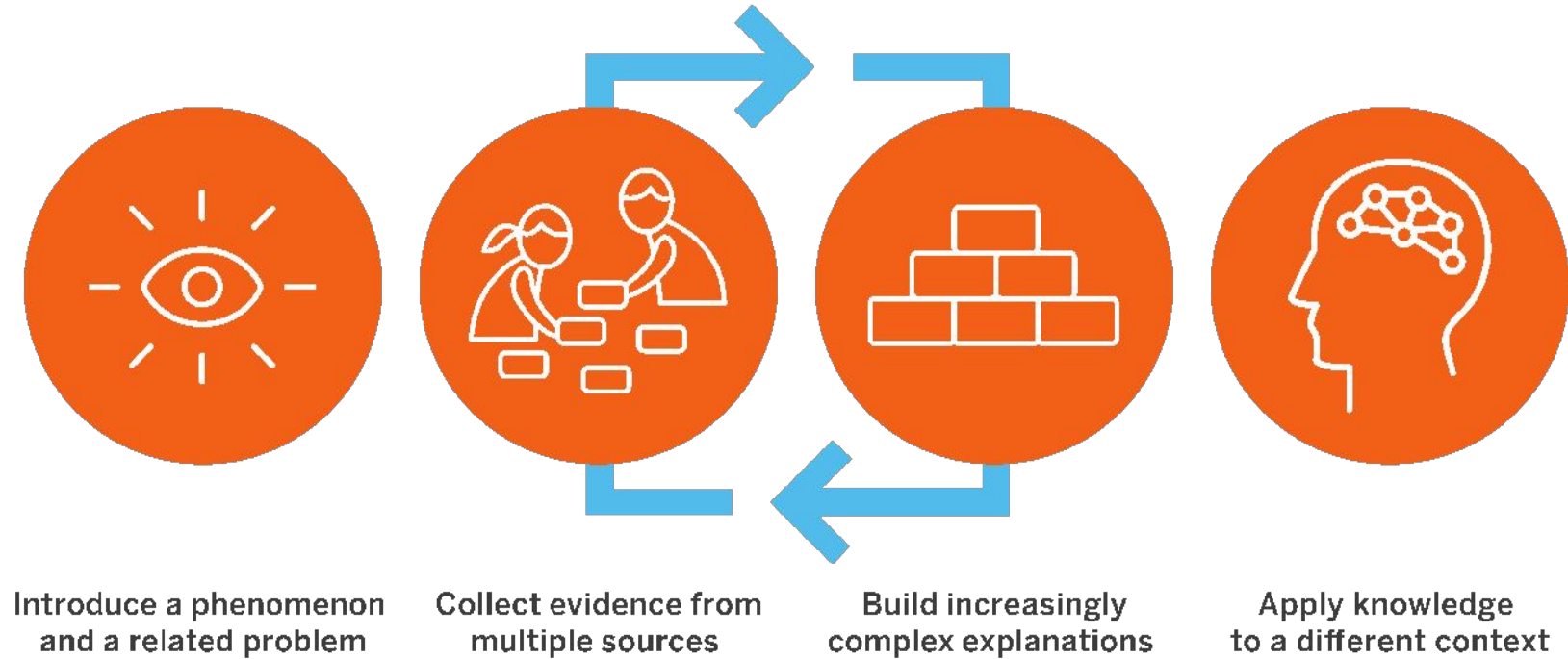
Crosscutting Concepts (CCC)

How students makes sense of phenomenon

- Patterns
- Cause and Effect
- Scale, Proportion and Quantity
- Systems and System Models
- Energy and Matter
- Structure and Function
- Stability and Change



Amplify Science approach



Multimodal learning

Gathering evidence from different sources

